PRACTICAL 1: DATA IMPORT & CLEANING IN EXCEL/PYTHON

Objective: To import a raw sales/HR dataset, inspect its structure, identify data quality issues, and produce a clean, analysis-ready dataset.

Business Scenario: You have joined the Analytics team of *Northwind Retail Pvt. Ltd.* The HR department has handed over an employee dataset (hr_data.csv) with 500 records. It contains missing values, duplicates, and suspicious income outliers.

Sample Dataset — hr_data.csv (sample of 12 rows shown):

EmployeeID	Name	Department	Age	Gender	MonthlyIncome	Joining Date	Attrition Flag
E001	Aarav Shah	sales	29	M	42000	12-03-2021	No
E002	Priya Nair	Sales	34	F		05-07-2019	No
E003	Rohit Verma	HR	41	M	55000	22-11-2018	Yes
E004	Rohit Verma	HR	41	M	55000	22-11-2018	Yes
E005	Sneha Iyer	finance	26	F	38000	15-01-2022	No
E006	Kunal Mehta	Finance	38	M	950000	30-06-2017	No
E007		Marketing	31	F	47000	09-09-2020	No

EmployeeID	Name	Department	Age	Gender	MonthlyIncome	Joining Date	Attrition Flag
E008	Divya Rao	sales	45	F	61000	14-02-2016	Yes
E009	Arjun Singh	IT	27	M	52000	2021/08/19	No
E010	Meena Kumar	Marketing	33	F	44000	03-12-2019	No
E011	Aditya Joshi	IT	29	M		25-04-2021	No
E012	Kavya Reddy	finance	36	F	49000	11-10-2018	No

Issues visible: blank MonthlyIncome (E002, E011), duplicate row (E003/E004), blank Name (E007), inconsistent Department casing ("sales" vs "Sales"), inconsistent date format (E009), and a suspicious outlier income (E006 = ₹9,50,000 vs typical ₹40,000-60,000 range).

Question 1

Import hr_data.csv into Excel and inspect how the data loads.

Procedure:

1. Open a new Excel workbook
2. Go to **Data** tab → click **Get Data → From File → From Text/CSV**
3. Browse to the location of hr_data.csv and select it → click **Import**
4. In the preview window, check the **File Origin** is set correctly (UTF-8) so special characters in names don't break
5. Confirm Excel has detected the correct **Delimiter** (Comma) in the preview pane
6. Verify each column header (EmployeeID, Name, Department, etc.) appears in its own column in the preview
7. Click **Load** to bring the data into a new worksheet
8. Rename the worksheet tab to "Raw_Data"

Expected Outcome: A worksheet with 500 rows (plus header) and 8 columns, matching the sample structure shown above.

Question 2

Inspect the dataset's shape, column data types, and a sample of records.

Procedure:

1. Click on cell A1, then press Ctrl + Shift + End to select the full used range — the Name Box and status bar will show the row/column count, giving you the dataset "shape" (e.g., 500 rows × 8 columns)
2. Click each column header one at a time and check the **Number Format** dropdown on the Home tab:
    - EmployeeID, Name, Department, Gender, AttritionFlag → should show "General" or "Text"
    - Age, MonthlyIncome → should show "General" or "Number"
    - JoiningDate → should show "Date"; if it shows "General" or "Text", this is a data type problem
3. Use **View tab → Freeze Panes → Freeze Top Row** so headers stay visible
4. Scroll through the first 5 rows and the last 5 rows manually to visually check for obvious irregularities
5. In a blank cell, type =COUNTA(A2:A501) to confirm the row count matches your expectation

Business Interpretation: Confirming row/column count ensures the file wasn't truncated during transfer. JoiningDate showing as "General"/Text (as seen with E009's 2021/08/19 format) will block any tenure calculation later, so this must be corrected before analysis.

Question 3

Identify missing values, duplicate rows, and potential outliers in MonthlyIncome.

Procedure:

Missing values:

1. Select the entire MonthlyIncome column (excluding header)
2. **Home tab → Conditional Formatting → Highlight Cells Rules → More Rules**
3. Choose **Format only cells that contain → Blanks** → set fill color to yellow → OK
4. Visually scan the column — blank cells are now highlighted (in the sample, E002 and E011)
5. Repeat the same check on the Name column (blank for E007)

Duplicate rows:

1. Select the full data range

2. **Home tab → Conditional Formatting → Highlight Duplicate Values** (using the full row combination, or just EmployeeID/Name as the key columns) to preview before deleting anything
3. In the sample, E003 and E004 are identical duplicate records
4. Once confirmed, this will be removed in Question 4 using **Data → Remove Duplicates**

Outliers in MonthlyIncome:

1. In a helper cell, calculate: =QUARTILE.INC(MonthlyIncome_range,1) → label this Q1
2. In another cell: =QUARTILE.INC(MonthlyIncome_range,3) → label this Q3
3. Calculate IQR: =Q3_cell - Q1_cell
4. Calculate Lower Bound: =Q1_cell - 1.5*IQR_cell
5. Calculate Upper Bound: =Q3_cell + 1.5*IQR_cell
6. Apply Conditional Formatting on MonthlyIncome with a **Custom Formula**: =OR(B2<Lower_Bound,B2>Upper_Bound) → highlight in red
7. In the sample data, E006's income of ₹9,50,000 will be flagged as an outlier against a typical Q1-Q3 band of roughly ₹42,000-₹55,000

Business Interpretation: Missing income values mean unrecorded payroll data. E006's outlier income may be a genuine senior-leadership salary or a data-entry error (an extra zero, i.e. ₹95,000 entered as ₹950,000) — verify with HR before deleting.

Question 4

Clean and standardize the dataset.

Procedure:

1. **Remove duplicates:** Select the full data range → **Data tab → Remove Duplicates** → tick all columns → OK (removes the E004 duplicate of E003)
2. **Fill missing income:** Build a small Pivot Table with Department in Rows and Average of MonthlyIncome in Values to get each department's median/average reference value. Then manually enter the department-level median into the blank cells for E002 (Sales) and E011 (IT), or use the array formula =MEDIAN(IF(Department_range="Sales",MonthlyIncome_range)) entered with Ctrl+Shift+Enter
3. **Handle missing Name:** Since E007's name is missing and cannot be reliably inferred, flag the row and either follow up with HR or exclude it from name-based analysis (do not delete the row, since income/department data are still valid)
4. **Standardize text columns:**
    - In a new helper column, use =PROPER(Department_cell) to convert "sales"/"Sales" both into "Sales"
    - Copy the helper column, then **Paste Special → Values** back over the original Department column
    - Delete the helper column
    - Repeat using =UPPER(Gender_cell) for the Gender column
5. **Fix JoiningDate format:** Select the JoiningDate column → **Data tab → Text to Columns** → choose **Delimited** → Next → Next → under Column Data Format select **Date: YMD** (to match E009's format) → Finish. For rows already in DD-MM-YYYY format, verify they did not get corrupted; fix manually if needed

6. Save the cleaned file as hr_data_cleaned.xlsx

Resulting cleaned sample:

EmployeeID	Name	Department	Age	Gender	MonthlyIncome	Joining Date	Attrition Flag
E001	Aarav Shah	Sales	29	M	42000	12-03-2021	No
E002	Priya Nair	Sales	34	F	46500	05-07-2019	No
E005	Sneha Iyer	Finance	26	F	38000	15-01-2022	No
E006	Kunal Mehta	Finance	38	M	950000	30-06-2017	No
E009	Arjun Singh	IT	27	M	52000	19-08-2021	No

Question 5

Document each cleaning step taken, in a log a manager could review.

Procedure: Create a new worksheet tab named "Cleaning Log" and record each action in a table.

Data Cleaning Log:

Step	Action Taken	Reason
1	Removed 1 duplicate row (E004, duplicate of E003)	Same EmployeeID/record appeared twice due to export error
2	Filled missing income for E002, E011 with department median	Avoids distortion from extreme values vs. mean

Step	Action Taken	Reason
3	Flagged E007 for missing Name; retained row	No reliable way to infer name, but other fields are valid
4	Standardized Department/Gender text casing	Prevents “sales” and “Sales” being treated as different categories
5	Converted JoiningDate to a single consistent date format	Enables tenure calculations later
6	Flagged E006 income (₹9,50,000) for HR verification	Possible data-entry error; not deleted pending confirmation

PRACTICAL 2: DESCRIPTIVE STATISTICS ON BUSINESS DATA

Objective: To compute and interpret central tendency, dispersion, and distribution-shape statistics on business data.

Business Scenario: Finance wants a statistical summary of MonthlyIncome across the company for the annual compensation review, using the cleaned hr_data_cleaned.xlsx from Practical 1.

Sample Dataset (MonthlyIncome column, 12 values used for illustration):

EmployeeID	MonthlyIncome
E001	42000
E002	46500
E003	55000
E005	38000
E006	95000
E008	61000
E009	52000
E010	44000
E011	48000
E012	49000
E013	53000
E014	41000

(Note: E006's outlier was corrected to ₹95,000 after HR confirmed the extra zero was a data-entry error.)

Question 1

Compute mean, median, and mode of MonthlyIncome.

Procedure:

1. Click an empty cell, label it "Mean" → enter =AVERAGE(MonthlyIncome_range)
2. Click the next cell, label it "Median" → enter =MEDIAN(MonthlyIncome_range)
3. Click the next cell, label it "Mode" → enter =MODE.SNGL(MonthlyIncome_range)
4. Format all result cells as currency (₹) using Home → Number Format → Currency

Worked Result (sample data): Mean ≈ ₹52,042 | Median ≈ ₹48,500 | Mode = N/A (no exact repeat in this small sample; on the full 500-row dataset a mode will emerge)

Interpretation: Mean is slightly above the median, suggesting a few higher earners (like E006) are pulling the average up. The median is the fairer figure to quote when communicating "typical" pay to the workforce.

Question 2

Compute variance, standard deviation, and range of MonthlyIncome.

Procedure:

1. Label a cell "Variance" → enter =VAR.S(MonthlyIncome_range)
2. Label a cell "Std Dev" → enter =STDEV.S(MonthlyIncome_range)
3. Label a cell "Range" → enter =MAX(MonthlyIncome_range)-MIN(MonthlyIncome_range)

Worked Result (sample data): Std Dev ≈ ₹15,200 | Range = ₹95,000 − ₹38,000 = ₹57,000

Interpretation: A standard deviation of roughly ₹15,200 around a mean of ₹52,000 indicates considerable pay variation across this small group — useful evidence if HR is investigating pay-equity concerns across departments.

Question 3

Compute quartiles and the Interquartile Range (IQR).

Procedure:

1. Label a cell "Q1" → enter =QUARTILE.INC(MonthlyIncome_range,1)
2. Label a cell "Q2 (Median)" → enter =QUARTILE.INC(MonthlyIncome_range,2)
3. Label a cell "Q3" → enter =QUARTILE.INC(MonthlyIncome_range,3)
4. Label a cell "IQR" → enter =Q3_cell - Q1_cell

Worked Result (sample data): Q1 ≈ ₹43,500 | Q3 ≈ ₹54,250 | IQR ≈ ₹10,750

Interpretation: The middle 50% of employees in this sample earn between roughly ₹43,500 and ₹54,250. A narrow IQR like this suggests most staff are clustered in a similar pay band, with E006 sitting well outside it.

Question 4

Compute skewness and kurtosis of MonthlyIncome.

Procedure:

1. Label a cell "Skewness" → enter =SKEW(MonthlyIncome_range)
2. Label a cell "Kurtosis" → enter =KURT(MonthlyIncome_range)

Worked Result (sample data): Skewness ≈ +1.6 (positive) | Kurtosis ≈ +2.1 (leptokurtic, i.e. peaked with heavier tails than normal)

Interpretation: Positive skewness confirms a long right tail caused by E006's high salary — typical of compensation data. The elevated kurtosis indicates income is tightly clustered around the median with one or two extreme outliers pulling the tail, rather than being smoothly spread out.

Question 5

Interpret all statistics together in a business summary.

Procedure: Compile all results computed above into one summary table on a new worksheet tab named "Descriptive Summary".

Statistical Summary Report:

Statistic	Value	Business Meaning
Mean	₹52,042	Average pay across sampled employees
Median	₹48,500	Pay of the "typical" employee
Std Dev	₹15,200	Degree of pay variation
IQR	₹10,750	Spread of the middle-earning 50%
Skewness	+1.6	A few high earners pulling distribution right

Conclusion: Median pay should be used in employee communications; mean and standard deviation are more useful internally for budgeting total payroll outlay, and the skew flags that compensation analysis should treat E006-type salaries separately from the typical pay band.

PRACTICAL 3: PIVOT TABLES AND CROSS-TABULATION

Objective: To summarize sales data by region and product-year combinations using pivot tables, and extract business insights.

Business Scenario: The Sales Director wants a regional and product-wise performance summary from sales_data.csv before the quarterly review.

Sample Dataset — sales_data.csv (sample of 14 rows shown):

OrderID	Region	Product	Year	Month	SalesAmount	AdSpend
S101	North	Electronics	2023	Jan	125000	18000
S102	North	Apparel	2023	Jan	64000	9000
S103	South	Electronics	2023	Jan	58000	7000
S104	East	Apparel	2023	Feb	71000	9500
S105	West	Electronics	2023	Feb	98000	14000
S106	North	Electronics	2024	Jan	142000	21000
S107	South	Apparel	2024	Jan	49000	6000
S108	South	Electronics	2024	Feb	51000	6500
S109	East	Electronics	2024	Feb	87000	12000
S110	North	Apparel	2024	Mar	69000	9800
S111	West	Apparel	2023	Mar	56000	7200
S112	West	Electronics	2024	Mar	102000	15500
S113	East	Apparel	2024	Apr	73000	9700
S114	South	Electronics	2023	Apr	53000	6800

Question 1

Build a pivot table showing total Sales by Region.

Procedure:

1. Click anywhere inside the data range
2. **Insert tab → PivotTable → New Worksheet → OK**
3. In the Field List on the right, drag **Region** into the **Rows** box
4. Drag **SalesAmount** into the **Values** box (Excel defaults to Sum — confirm it says "Sum of SalesAmount")
5. Rename the pivot sheet tab to "Sales by Region"

Expected Output (based on sample data):

Region	Sum of SalesAmount
East	231,000
North	400,000
South	211,000
West	256,000

Question 2

Build a pivot table showing Sales by Product × Year.

Procedure:

1. Insert a second PivotTable from the same source data range
2. Drag **Product** to **Rows**
3. Drag **Year** to **Columns**
4. Drag **SalesAmount** to **Values** (Sum)
5. Rename the pivot sheet tab "Sales by Product-Year"

Expected Output (based on sample data):

Product	2023	2024
Apparel	191,000	191,000
Electronics	281,000	382,000

Question 3

Add slicers and filters to the Region pivot table.

Procedure:

1. Click inside the "Sales by Region" pivot table

2. **PivotTable Analyze tab → Insert Slicer**
3. Tick **Product** in the dialog → OK
4. A slicer box with buttons "Electronics" and "Apparel" appears — drag it to a clear area next to the pivot table
5. Click "Electronics" on the slicer to confirm the pivot table updates to show only electronics sales by region

Question 4

Create a pivot chart from the Region pivot table.

Procedure:

1. Click inside the "Sales by Region" pivot table
2. **PivotTable Analyze tab → PivotChart**
3. Select **Clustered Column Chart** → OK
4. Click the chart title placeholder and type "Regional Sales Performance"
5. Right-click the Product slicer → **Report Connections** → tick the new PivotChart's underlying pivot table so the slicer also filters the chart

Question 5

Derive 3 business insights from the pivot tables created above.

Business Insights:

1. **Region concentration:** North contributes the highest total sales (₹400,000 in the sample) — priority market for inventory allocation.
2. **Product-Year trend:** Electronics grew from ₹281,000 (2023) to ₹382,000 (2024), while Apparel stayed flat at ₹191,000 both years — apparel line may need refreshing.
3. **Underperforming combination:** South's electronics sales (₹58,000 + ₹51,000 + ₹53,000 ≈ ₹162,000 across the sample) are the lowest among regions for that category — signals a distribution or marketing gap in South.

PRACTICAL 4: DATA VISUALIZATION — CHARTS AND PLOTS

Objective: To create and interpret standard business charts for communicating data patterns.

Business Scenario: Marketing needs a visual report comparing product performance, trends, and distribution from sales_data.csv for the board presentation. (Using the same sample dataset as Practical 3.)

Question 1

Create a bar/column chart comparing total sales by Region.

Procedure:

1. Use the Region summary table from Practical 3, Question 1 (or build one with =SUMIF(Region_range,"North",SalesAmount_range) for each region)
2. Select the two-column summary table (Region, Sum of SalesAmount)
3. **Insert tab → Charts group → Clustered Column Chart**
4. Click **Chart Title** → type "Total Sales by Region"
5. Click **+ (Chart Elements) → Axis Titles** → label Y-axis "Sales Amount (₹)"
6. **Chart Design → Change Colors** to apply a consistent blue theme

Question 2

Create a line chart showing monthly sales trend.

Procedure:

1. Build a Month vs. Total Sales summary table using a Pivot Table (Month in Rows, Sum of SalesAmount in Values) — sort chronologically (Jan, Feb, Mar, Apr)
2. Select the table → **Insert → Line Chart with Markers**
3. Title the chart "Monthly Sales Trend"
4. Label the X-axis "Month" and Y-axis "Sales Amount (₹)"

Expected trend table (sample data):

Month	Sum of SalesAmount
Jan	438,000
Feb	307,000
Mar	227,000

Month	Sum of SalesAmount
Apr	126,000

Question 3

Create a pie/donut chart showing each product's share of total sales.

Procedure:

1. Build a Product vs. Total Sales summary table (Electronics: 663,000 | Apparel: 382,000 from the sample)
2. Select the table → **Insert → Pie Chart** (or **Doughnut Chart** for a donut style)
3. Right-click the chart → **Add Data Labels → More Options** → tick **Percentage**, untick Value
4. Title the chart "Product Share of Total Sales"

Question 4

Create a scatter plot for the relationship between Advertising Spend and Sales.

Procedure:

1. Select the AdSpend and SalesAmount columns together (use Ctrl to select both non-adjacent columns if needed)
2. **Insert → Scatter (X,Y) Chart**
3. Click + → **Axis Titles** → label X-axis "Ad Spend (₹)" and Y-axis "Sales Amount (₹)"
4. Right-click any data point → **Add Trendline → Linear** → tick **Display R-squared value on chart**

Question 5

Create a boxplot for distribution and outliers in Sales by Region.

Procedure:

1. Select the Region and SalesAmount columns
2. **Insert → Insert Statistic Chart → Box and Whisker**
3. Excel automatically groups the data by Region and plots quartiles, median line, and any outlier points
4. Title the chart "Sales Distribution by Region"

Question 6

Explain when to use each chart type in a business context.

Chart Type	Best Used For
Bar/Column	Comparing totals across discrete categories (e.g., region, product)
Line	Showing trends over a continuous time period (e.g., monthly sales)
Pie/Donut	Showing each category's share of a whole, when categories are few (≤6)
Scatter	Studying relationship/correlation between two numeric variables
Boxplot	Comparing distribution spread and spotting outliers across groups

PRACTICAL 5: EXPLORATORY DATA ANALYSIS (EDA)

Objective: To systematically explore a dataset using univariate, bivariate, and grouped analysis to surface actionable insights.

Business Scenario: The strategy team wants a full exploratory report on sales_data.csv before deciding next quarter's product investment priorities. (Continuing with the Practical 3/4 sample dataset.)

Question 1

Perform univariate analysis on each numeric column.

Procedure:

1. Enable the Analysis ToolPak if not already active: **File → Options → Add-ins → Manage: Excel Add-ins → Go → tick Analysis ToolPak → OK**
2. **Data tab → Data Analysis → Descriptive Statistics**
3. Set Input Range = SalesAmount column → tick **Summary Statistics** → Output to a new worksheet → OK
4. Repeat for AdSpend column
5. For each numeric column, also create a histogram: select the column → **Insert → Insert Statistic Chart → Histogram**

Interpretation: Univariate analysis reveals the spread and shape of each individual variable — whether sales are normally distributed or skewed by a few large transactions (in the sample, Electronics orders like S106 at ₹142,000 sit well above the typical ₹50,000-₹70,000 range).

Question 2

Perform bivariate analysis between key variable pairs (e.g., AdSpend vs SalesAmount).

Procedure:

1. Create the scatter plot from Practical 4, Question 4
2. Compute the correlation coefficient in a blank cell: =CORREL(AdSpend_range, SalesAmount_range)
3. Repeat for another pair if relevant data is available (e.g., Month-index vs SalesAmount to check seasonality)

Worked Result (sample data): Correlation ≈ 0.93 — a strong positive relationship between Ad Spend and Sales.

Question 3

Build a correlation matrix and heatmap for all numeric variables.

Procedure:

1. **Data tab → Data Analysis → Correlation**
2. Set Input Range to cover SalesAmount and AdSpend (and any other numeric columns) → tick **Labels in First Row** → Output to new location → OK
3. Excel produces a correlation matrix table
4. Select the matrix cells → **Home → Conditional Formatting → Color Scales** → choose a Red-Yellow-Green scale to convert it into a heatmap

Question 4

Generate group-wise summaries (e.g., average sales per Region and Product).

Procedure:

1. Build a PivotTable with **Region** and **Product** both in **Rows** (Region first, Product nested under it)
2. Drag **SalesAmount** into Values three separate times
3. For each of the three instances, right-click → **Value Field Settings** → set one to **Average**, one to **Sum**, one to **Count**
4. Rename headers to "Avg Sales", "Total Sales", "Order Count"

Question 5

Identify and document 5 business insights in an EDA report.

EDA Insight Report:

1. **Strong AdSpend–Sales correlation (r ≈ 0.93):** Higher marketing investment is strongly associated with higher sales, supporting continued ad budget allocation.
2. **Skewed sales distribution:** A small number of large Electronics transactions inflate average sales — median is a more reliable benchmark than mean.
3. **Regional disparity:** North consistently outperforms other regions across nearly every product category.
4. **Seasonal pattern:** Sales taper from ₹438,000 in January to ₹126,000 in April in the sample — suggesting a front-loaded demand cycle worth planning inventory around.
5. **Weak performer identified:** South shows the lowest electronics sales despite reasonable ad spend, indicating a possible distribution or market-fit gap rather than a marketing budget gap.

PRACTICAL 6: CORRELATION AND SIMPLE LINEAR REGRESSION

Objective: To quantify the relationship between Advertising Spend and Sales, and build a predictive regression model.

Business Scenario: The Marketing Director wants to know: "If we increase ad spend, how much extra sales can we expect?" Using the AdSpend/SalesAmount columns from sales_data.csv (same sample as Practical 3).

Question 1

Compute the Pearson correlation coefficient between AdSpend and SalesAmount.

Procedure:

1. In a blank cell: =CORREL(AdSpend_range, SalesAmount_range)

Worked Result: $r \approx 0.93$

Interpretation: A correlation close to +1 indicates a strong positive linear relationship — as ad spend rises, sales tend to rise too.

Question 2

Plot a scatter plot with a fitted regression line.

Procedure:

1. Select AdSpend and SalesAmount columns → **Insert → Scatter Chart**
2. Right-click any data point → **Add Trendline**
3. Choose **Linear** → tick **Display Equation on Chart** and **Display R-squared value on chart**

Expected Equation (sample data, approx.): $y \approx 5.9x + 14{,}200$, $R^2 \approx 0.87$

Question 3

Fit a simple linear regression model using the Data Analysis tool.

Procedure:

1. **Data tab → Data Analysis → Regression**
2. Set **Input Y Range** = SalesAmount column

3. Set **Input X Range** = AdSpend column
4. Tick **Labels** if headers are included in the selected range → OK
5. Excel outputs a Regression Statistics table including Intercept, X Variable coefficient (Slope), and R Square

Question 4

Interpret the intercept, slope, and R^2 value of the model.

Procedure: Read the values directly from the Regression output table generated in Question 3.

Interpretation (using sample figures):

- **Intercept (≈ ₹14,200):** Expected sales when ad spend is zero (baseline/organic sales)
- **Slope (≈ 5.9):** For every additional ₹1 spent on advertising, sales increase by approximately ₹5.9 on average
- **R^2 (≈ 0.87):** About 87% of the variation in sales is explained by ad spend alone; the remainder is due to other factors (seasonality, pricing, region, competition)

Question 5

Predict expected sales for new ad-spend values of ₹20,000 and ₹30,000.

Procedure:

1. In a cell: =Intercept_cell + Slope_cell*20000
2. In another cell: =Intercept_cell + Slope_cell*30000

Worked Result (sample figures): At ₹20,000 ad spend → predicted sales ≈ ₹1,32,200. At ₹30,000 ad spend → predicted sales ≈ ₹1,91,200.

Business Recommendation: Since predicted incremental sales (~₹59,000 for an extra ₹10,000 spend) substantially exceed the additional cost, Marketing has a data-backed case to request a higher ad budget next quarter.

PRACTICAL 7: FORECASTING USING MOVING AVERAGE / EXPONENTIAL SMOOTHING

Objective: To forecast future monthly sales and evaluate forecast accuracy.

Business Scenario: Inventory Planning needs a sales forecast for next quarter based on historical monthly sales (monthly_sales.csv).

Sample Dataset — monthly_sales.csv:

Month	Sales
Jan-2024	210000
Feb-2024	198000
Mar-2024	225000
Apr-2024	240000
May-2024	232000
Jun-2024	251000
Jul-2024	264000
Aug-2024	258000
Sep-2024	270000
Oct-2024	285000
Nov-2024	298000
Dec-2024	310000

Question 1

Plot the monthly sales time series.

Procedure:

1. Select the Month and Sales columns
2. **Insert → Line Chart**
3. Title the chart "Monthly Sales Time Series"
4. Label the Y-axis "Sales (₹)"

Question 2

Apply 3-month and 6-month moving averages.

Procedure:

1. **Data tab → Data Analysis → Moving Average**
2. Set Input Range = Sales column, Interval = 3, Output Range = new column → OK
3. Repeat with Interval = 6 in another new column
4. Select Sales, MA-3, and MA-6 columns together → **Insert → Line Chart** to compare all three on one chart

Worked sample (MA-3, first calculable value at Mar): MA-3 for Mar = AVERAGE(210000,198000,225000) = ₹211,000

Question 3

Apply Simple Exponential Smoothing.

Procedure:

1. **Data tab → Data Analysis → Exponential Smoothing**
2. Set Input Range = Sales column
3. Set Damping factor = 0.7 (this corresponds to a smoothing level $\alpha = 1 - 0.7 = 0.3$)
4. Output Range = new column → OK
5. Add this output as a third line on the comparison chart from Question 2

Question 4

Compute MAE and RMSE forecast errors for each method.

Procedure:

1. Create an "Error" column for each method: =Actual_cell - Forecast_cell
2. MAE: =AVERAGE(ABS(error_range)) — for array entry in older Excel, use Ctrl+Shift+Enter
3. RMSE: =SQRT(AVERAGE(error_range^2)) entered as an array formula in older Excel, or directly in Excel 365 with dynamic arrays
4. Repeat for MA-3, MA-6, and SES error columns, then compare the three MAE/RMSE results side by sid

Question 5

Compare the methods and recommend the best one.

Comparison Table (template to fill from your computed values):

Method	MAE	RMSE	Responsiveness to Recent Change
3-Month MA	[value]	[value]	Moderate
6-Month MA	[value]	[value]	Low (smoother, slower to react)
Simple Exponential Smoothing	[value]	[value]	High (weights recent data more)

Recommendation: The method with the lowest MAE/RMSE should be adopted. Given the sample shows a steady upward trend (Jan ₹210,000 → Dec ₹310,000), Exponential Smoothing typically tracks this rising trend faster than a 6-month MA, which lags behind sudden changes.

PRACTICAL 8: CUSTOMER SEGMENTATION USING CLUSTERING

Objective: To segment customers into meaningful groups based on income and spending behaviour.

Business Scenario: Marketing wants customers grouped into segments based on customer_data.csv to design targeted campaigns.

Sample Dataset — customer_data.csv:

CustomerID	AnnualIncome (₹'000)	SpendingScore (1-100)
C001	250	82
C002	270	88
C003	180	25
C004	190	18
C005	95	75
C006	88	80
C007	60	22
C008	55	30
C009	260	90
C010	75	20
C011	100	78
C012	185	22

Question 1

Load the customer dataset and inspect Income and Spending columns.

Procedure:

1. Open customer_data.csv in Excel
2. Use =AVERAGE(), =MIN(), =MAX() on both AnnualIncome and SpendingScore to get a quick sense of range

Worked Result (sample): Income range ₹55,000–₹270,000; Spending Score range 18–90.

Question 2

Standardize the features before clustering.

Procedure:

1. In a new column, compute the Z-score for income: =(AnnualIncome_cell - AVERAGE(AnnualIncome_range)) / STDEV.S(AnnualIncome_range)
2. In another new column, compute the Z-score for spending: =(SpendingScore_cell - AVERAGE(SpendingScore_range)) / STDEV.S(SpendingScore_range)
3. Apply both formulas down the full customer list

Why standardize? Income (tens of thousands) and Spending Score (1-100) are on very different scales; without standardization, income differences alone would dominate the distance calculation and distort the clusters.

Question 3

Apply k-means clustering with k = 3, 4, and 5 (manual Excel approach).

Procedure:

1. Choose k initial "center" points by picking k random customer rows from the standardized data as starting centers (e.g., for k=3: C001, C007, C005)
2. For each customer, calculate distance to each center: =SQRT((Z_Income-CenterZ_Income)^2+(Z_Spend-CenterZ_Spend)^2)
3. Use =MIN() across the distance columns to find the smallest distance, then =MATCH() to identify which center that corresponds to — this becomes the customer's assigned cluster
4. Recalculate each center as the average Z-Income/Z-Spending of its currently assigned members
5. Repeat steps 2–4 two to three times until cluster assignments stop changing
6. Repeat the entire process separately for k=4 and k=5

Note: This manual approach builds intuition for how k-means works; for the full 500-row dataset, the same logic is automated faster in Power BI's clustering visual.

Question 4

Use the elbow method to pick the optimal k.

Procedure:

1. For each k value tested (1 through 6), sum the squared distance of every point to its assigned center (Within-Cluster Sum of Squares)

2. Build a small table: k in one column, Total Distance in the next
3. Select this table → **Insert → Line Chart**
4. Identify the "elbow" — the point where the line's downward slope flattens noticeably

Interpretation: Based on the sample's natural grouping (high-income/high-spend, low-income/high-spend, mid-income/low-spend), k=3 or k=4 is likely to be the elbow point.

Question 5

Profile and name each segment.

Procedure:

1. Once the final k is chosen, build a Pivot Table with Cluster in Rows and Average of AnnualIncome / Average of SpendingScore in Values
2. Assign a descriptive name to each segment based on its profile

Segment Naming (based on sample data, k=3):

Segment	Avg Income	Avg Spending	Suggested Name
0 (C001,C002,C009)	₹260,000	87	"Premium Loyalists"
1 (C003,C004,C012)	₹185,000	22	"Cautious Affluent"
2 (C005,C006,C011)	₹94,000	78	"Value Seekers"

Question 6

Visualize the clusters.

Procedure:

1. Select Income, Spending, and Cluster columns
2. **Insert → Scatter Chart**
3. To color-code by cluster, filter the data by each cluster number one at a time and add each as a separate chart series (right-click chart → Select Data → Add), or use Power BI's scatter visual with Cluster as the Legend field for automatic coloring

Business Recommendation: "Premium Loyalists" should receive loyalty rewards to retain them; "Value Seekers" respond well to discount-driven campaigns; "Cautious Affluent" customers may need targeted upselling to convert income into spend.

PRACTICAL 9: A/B TESTING ON MARKETING DATA

Objective: To statistically test whether Campaign B outperforms Campaign A, and make a data-backed recommendation.

Business Scenario: Two email campaign versions were tested — Campaign A (control) and Campaign B (new design). Leadership wants to know which to roll out.

Sample Dataset — ab_test_data.csv (sample of 16 rows shown; Converted: 1 = Yes, 0 = No):

UserID	Group	Converted
U001	A	0
U002	A	1
U003	A	0
U004	A	0
U005	A	1
U006	A	0
U007	A	0
U008	A	1
U009	B	1
U010	B	1
U011	B	0
U012	B	1
U013	B	1
U014	B	0
U015	B	1
U016	B	1

(Full dataset would have hundreds of users per group; this sample illustrates the method.)

Question 1

Load the campaign data and separate Group A and Group B.

Procedure:

1. Open ab_test_data.csv
2. Apply **Data → Filter** on the Group column, or build a PivotTable with Group in Rows to view A and B separately

Question 2

Compute the conversion rate for both groups.

Procedure:

1. For Group A: =AVERAGEIF(Group_range,"A",Converted_range)
2. For Group B: =AVERAGEIF(Group_range,"B",Converted_range)
3. Format both as percentages

Worked Result (sample data): Group A: 3/8 = 37.5% | Group B: 6/8 = 75%

Question 3

Perform a two-proportion test to compare the conversion rates.

Procedure:

1. Count successes: =COUNTIFS(Group_range,"A",Converted_range,1) and the same for "B"
2. Count totals: =COUNTIF(Group_range,"A") and the same for "B"
3. Compute pooled proportion: $\hat{p}$ = (successes_A + successes_B) / (n_A + n_B)
4. Compute standard error: SE = SQRT($\hat{p}$*(1-$\hat{p}$)*(1/n_A + 1/n_B))
5. Compute Z-statistic: Z = (p_B - p_A) / SE
6. Convert Z to a two-tailed p-value: =2*(1-NORM.S.DIST(ABS(Z_cell),TRUE))

Question 4

Compute and interpret the p-value at a 95% confidence level.

Procedure: Compare the resulting p-value against $\alpha = 0.05$.

Interpretation: A p-value below 0.05 means the observed difference in conversion rates (37.5% vs 75% in the sample) is unlikely to be due to random chance alone, giving confidence that Campaign B genuinely drives more conversions. (Note: with only 8 users per group, real statistical power requires the full dataset — this sample is for illustrating the calculation method.)

Question 5

Make a business recommendation based on the test result.

Recommendation:

If p-value < 0.05 and Campaign B's conversion rate is higher: *"Campaign B should be rolled out to the full customer base — the uplift is statistically significant and not attributable to chance."*

If p-value ≥ 0.05: *"There is insufficient statistical evidence to declare Campaign B superior. Recommend extending the test duration or increasing sample size before making a rollout decision."*

PRACTICAL 10: MINI CASE STUDY & DASHBOARD

Objective: To independently apply the full analytics workflow — cleaning, EDA, modeling, and dashboarding — to a real business problem.

Business Scenario: *Northwind Retail*'s leadership has noticed a sales decline over the last two quarters and wants the Analytics team to investigate and present a dashboard at the next strategy meeting.

Sample Dataset — `case_study_data.csv` (sample of 10 rows shown):

Date	Region	Product	SalesAmount	AdSpend	DiscountPercent	CompetitorPriceIndex
01-01-2024	North	Electronics	142000	21000	5	98
01-02-2024	North	Electronics	138000	19500	5	99
01-03-2024	North	Electronics	129000	17000	8	102
01-04-2024	North	Electronics	118000	15000	8	105
01-05-2024	North	Electronics	109000	13000	10	108
01-06-2024	North	Electronics	101000	11500	10	110
01-01-2024	South	Apparel	49000	6000	5	97
01-02-2024	South	Apparel	47000	5800	5	98
01-03-2024	South	Apparel	44000	5200	7	100

Date	Region	Product	SalesAmount	AdSpend	DiscountPercent	CompetitorPriceIndex
01-04-2024	South	Apparel	40000	4500	7	103

Pattern visible: as AdSpend declines and CompetitorPriceIndex rises month over month, SalesAmount steadily falls — this is the core relationship students should uncover.

Question 1

Pick a business problem and define it clearly.

Problem Definition:
"Quarterly sales for Northwind Retail have declined by 12% over the last two quarters. The objective is to identify the key drivers of this decline using available transaction, customer, and marketing data, and recommend corrective action."

Question 2

Clean and prepare the relevant dataset(s).

Procedure: Apply the full cleaning checklist from Practical 1:

1. Remove duplicate rows (Data → Remove Duplicates)
2. Handle missing values using category-wise medians
3. Fix data types, especially the Date column (Text to Columns → Date)
4. Standardize text fields (Region, Product) using PROPER()
5. Save the cleaned file as case_study_cleaned.xlsx

Question 3

Perform EDA and apply at least one modeling technique to investigate the problem.

Procedure:

1. Build a monthly sales trend line chart (Date vs. SalesAmount) — as in Practical 7, Question 1 — to visually confirm the decline shown in the sample (North Electronics: ₹142,000 in Jan down to ₹101,000 in Jun)
2. Build a correlation matrix between SalesAmount, AdSpend, DiscountPercent, and CompetitorPriceIndex using **Data Analysis → Correlation** (as in Practical 5, Question 3)
3. Identify which variable has the strongest relationship with the decline

Finding (based on sample pattern): SalesAmount shows a strong positive correlation with AdSpend (as ad spend dropped from ₹21,000 to ₹11,500, sales fell correspondingly) and a strong negative correlation with CompetitorPriceIndex (as competitor pricing index rose from 98 to 110, Northwind's sales fell) — suggesting both reduced marketing visibility and pricing pressure are contributing factors.

Question 4

Build a simple dashboard in Excel or Power BI summarizing the key metrics.

Procedure:

1. Create a new worksheet tab named "Dashboard"
2. Add **KPI Cards**: Total Sales, % Change vs. Previous Quarter, Average Discount % (use large merged cells with bold formulas linked to summary calculations, e.g. =(ThisQtr-LastQtr)/LastQtr)
3. Insert the **Monthly Sales Trend** line chart from Question 3, and use a shaded data series or text box annotation to highlight the declining months
4. Insert a **Bar Chart** of Sales by Region/Product (as in Practical 4, Question 1)
5. Insert a **Pivot Table with Slicer** allowing filtering by Region, Product, and Month
6. Arrange all elements on one screen using consistent fonts, a single color theme (navy/blue), and aligned grid spacing for a board-ready look

Question 5

Present findings and recommendations in a one-page summary.

Executive Summary:

Problem: 12% sales decline over the last two quarters.

Key Findings:

1. Ad spend reduction from ₹21,000 to ₹11,500 (North Electronics) correlates directly with the sharpest drop in sales.
2. South region's Apparel sales also declined steadily (₹49,000 → ₹40,000) alongside rising discount percentages, suggesting discounting alone isn't recovering volume.
3. Competitor pricing pressure intensified across the period (Competitor Price Index rising from ~97-98 to 103-110), especially affecting Electronics.

Recommendations:

1. Restore advertising budget to prior levels, particularly in North Electronics where the AdSpend-Sales link is strongest.
2. Conduct a pricing review for Electronics to remain competitive against the rising Competitor Price Index.

3. Launch a targeted regional campaign for South using insights from the customer segmentation analysis (Practical 8) rather than relying on discounting alone.

Conclusion: This case study demonstrates the full analytics lifecycle — from raw data to a stakeholder-ready dashboard — equipping students with an end-to-end business analytics workflow applicable to real organizational decision-making.